Emergency Vehicle Detection

Accelerating Smart City Response with YOLOv8

Deep Learning | Computer Vision | Object Detection

Presented By

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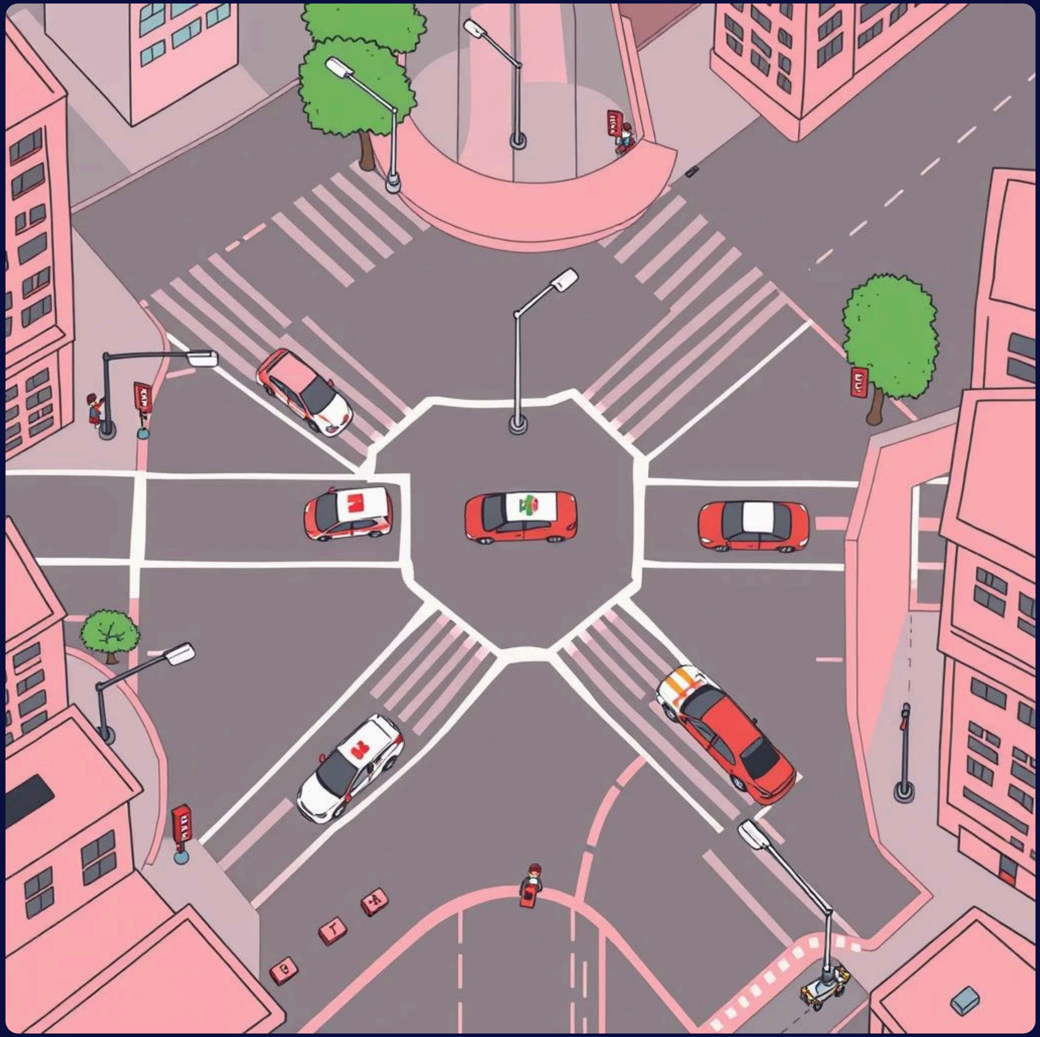
The Challenge: Time is Critical

Every second counts when emergency vehicles navigate congested urban environments. Timely detection is paramount for:

- Optimized traffic signal preemption
- Enhanced smart city infrastructure
- Improved emergency response times

Our goal: Develop a robust, real-time system to **accurately classify vehicles as emergency or non-emergency** within live traffic video streams.



Building the Foundation: Dataset Curation

A strong model begins with diverse and representative data. We meticulously collected and categorized images to ensure comprehensive coverage.

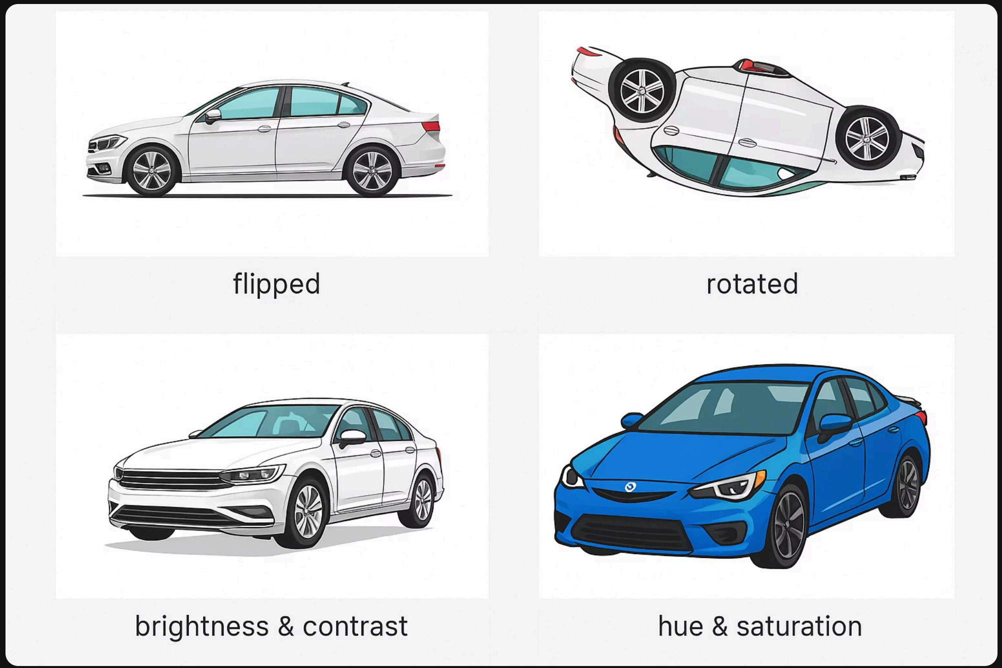
Emergency Vehicles	Non-Emergency Vehicles
• Ambulance: **269** images • Emergency Vehicles (general): **84** images • Rescue Vehicles: **11** images	• Casual Cars: **32** images • Pickup Trucks: **47** images • Sedans: **53** images • Vans: **53** images

Total Classes in Training: 7 (Mapped to 2 primary classes: Emergency & Non-Emergency)

Data Augmentation: Expanding Horizons

To prevent overfitting and enhance model robustness, we employed **Albumentations** for extensive data augmentation. This technique significantly expanded our dataset, making the model more resilient to real-world variations.

- Horizontal & Vertical Flips
- Random Rotations
- Brightness & Contrast Adjustments
- Hue & Saturation Variations



The final dataset is balanced and enriched, ensuring optimal generalization across diverse traffic scenarios.

Model Training: YOLOv8s Configuration

We selected **YOLOv8s** (small version) for its balance of efficiency and accuracy, crucial for real-time deployment.

1

Model Architecture

YOLOv8s: Optimized for speed and performance.

2

Training Epochs

75 Epochs: Iterative learning for refined feature extraction.

3

Image Resolution

960x960: High-resolution input for precise detection.

4

Optimizer & LR

AdamW (0.001): Efficient convergence and learning stability.

5

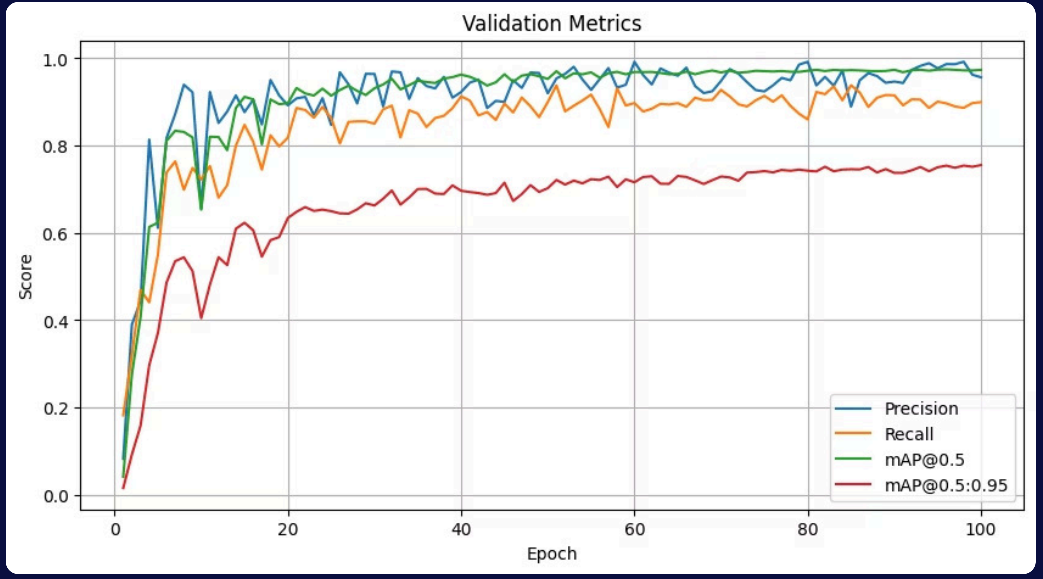
Output Classes

['emergency', 'non_emergency']: Binary classification for clarity.

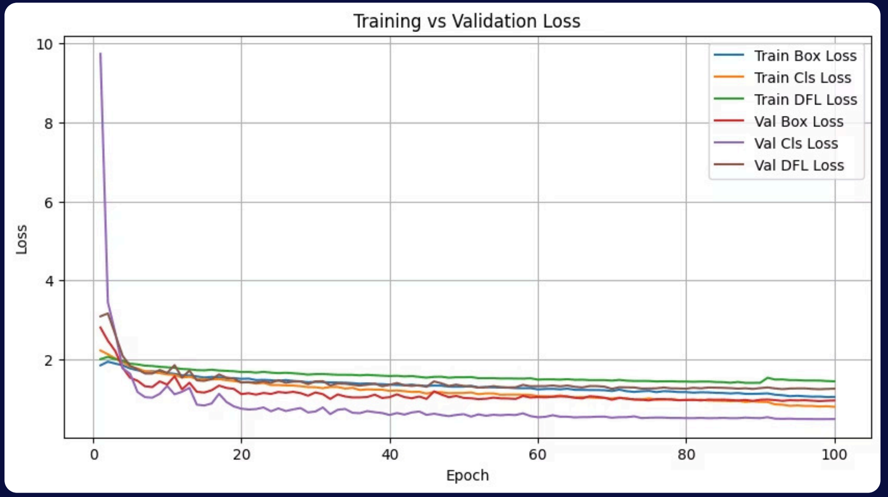
Exceptional Results Achieved

Our YOLOv8s model demonstrated outstanding performance, validating its capability for real-world application.

- **Precision & Recall:** Significant increases, minimizing false positives and false negatives.
- **mAP@0.5: 0.95**
  This indicates high accuracy in object detection, classifying vehicles correctly and localizing them precisely.



The loss curve demonstrates stable training, with both training and validation loss converging effectively. The metrics curve confirms the model's consistent performance improvement over epochs.



Real-Time Deployment: Vision in Action

The trained model was deployed on a live traffic video feed, providing instantaneous detection and classification.

Emergency Vehicles

Highlighted with a red bounding box.

Non-Emergency Vehicles

Marked with a green bounding box.

This visual feedback confirms accurate, real-time identification, enabling immediate action.

Key Takeaways from the Capstone

This project provided invaluable experience across the entire Deep Learning pipeline, focusing on a critical real-world problem.

End-to-End Workflow Mastery

From meticulous dataset preparation and augmentation to robust training, evaluation, and seamless deployment – I gained hands-on experience with every critical stage.

Real-World Application Impact

Successfully showcased the practical utility of Computer Vision by developing an effective system for detecting emergency vehicles in live traffic videos, directly addressing a critical urban challenge.

Strengthened Core Skills

Solidified my foundation in advanced Computer Vision and Deep Learning techniques, preparing me for complex challenges in AI and smart city development.

Shaping Smarter Cities: A Call to Action

- **Emergency vehicles are often stuck in traffic** due to a lack of priority signaling, hindering crucial response times.
- **Manual intervention is not scalable** in dense, rapidly growing urban environments.
- **A real-time, automated detection system** is essential to drastically improve emergency response and overall urban efficiency.

This project is a significant step towards creating smarter, more responsive cities and saving lives.

Thank You

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